

["shade": false] is missing from potted cross models, resulting in most potted plants appearing darker than they should

MC-129826 Rendering

Status	Resolved / Fixed
Affected	Minecraft 1.12.2, Minecraft 18w20c, Minecraft 18w22a, Minecraft 18w22c, Minecraft 1.13-pre5, Minecraft 1.13.1 (+55 more)
Fixed in	24w33a
Reported	2018-05-17

STEPS TO REPRODUCE

1. Launch Minecraft in a version from 1.8 onward.
2. Create a new Creative world with default settings.
3. Obtain a flower pot and a plant that can be placed in it.
4. Place the flower pot and place the plant in the flower pot.
5. Place the same type of plant outside the flower pot.
6. Observe that the plant in the flower pot appears noticeably darker than the same plant placed outside the flower pot.

OBSERVED BEHAVIOUR

Actual behavior

Since snapshot 14w25a, most plants rendered inside flower pots use cross-model elements that retain the default directional shading. The two flat planes in `flower_pot_cross.json` and `tinted_flower_pot_cross.json` therefore receive lighting based on their face directions, causing some faces to appear darker than others.

As a result, potted flowers, mushrooms, and other plants that use these cross models appear noticeably darker than the equivalent plants placed directly in the world. The darkening is visible from certain viewing directions and affects both normal and tinted potted cross models. This behavior has persisted from 1.8 through the current versions.

Potted plants with dedicated models, such as the mangrove propagule introduced in 22w11a, are not affected by this particular cross-model shading issue.

EXPECTED BEHAVIOUR

Plants displayed in flower pots should appear with consistent, properly lit colors and brightness, matching the appearance of the same plants placed directly in the world. Their flat leaves, flowers, grass, and mushrooms should not become noticeably darker based on their orientation, and tinted potted plants should receive the same correct rendering without unintended directional shading.

ENVIRONMENT

Minecraft Java Edition Versions: 1.12.2, 18w20c, 18w22a, 18w22c, 1.13-pre5, 1.13.1, 1.13.2, 18w43a, 18w43b, 18w43c, 18w47b, 18w48a, 18w48b, 19w04b, 19w05a, 19w12b, 19w13b, 19w14a, 19w14b, 1.14.4, 19w45a, 1.15.2, 20w15a, 1.16.3, 1.16.4, 20w51a, 21w03a, 21w05b, 21w06a, 21w07a, 21w10a, 21w11a, 21w13a, 21w14a, 21w16a, 1.17 Pre-release 1, 1.17 Pre-release 2, 1.17 Pre-release 3, 1.17, 1.17.1 Pre-release 1, 1.17.1, 21w39a, 21w40a, 21w42a, 1.18 Pre-release 1, 1.18, 1.18.1 Pre-release 1, 1.18.1, 22w06a, 1.18.2, 22w11a, 22w15a, 1.19 Pre-release 1, 1.19, 1.19.1 Pre-release 3, 1.19.2, 1.19.3, 23w04a, 23w18a, 1.20.1, 1.20.4 (includes snapshots)

ORIGINAL DESCRIPTION

The bug

When blocks were being converted over to using model files in 1.8, several of these conversions were incomplete and flawed, resulting in a multitude of issues. Many of these were resolved before the release of 1.8, however this specific issue was not.

In 14w25a, models involving flat 2D planes had those planes be rendered darker depending on their directions. This affected blocks such as grass, flowers and mushrooms, and was ultimately fixed (

MC-57159

Resolved

). However, the same version also caused plants in flower pots to appear darker for the same reason, and this was not fixed.

As a result, plants which are placed in flower pots appear noticeably darker than those same plants when placed in the world (outside of a flower pot), which has been the case since 14w25a.

A comparison of the appearances of potted objects can be found below:

1.4.2 - 1.7.10 (expected rendering)

1.8 - present (incorrect rendering)

How to fix

This is ultimately incredibly easy to fix - the two texture plane elements in "flower_pot_cross.json" and its tinted version "tinted_flower_pot_cross.json" simply need a

"shade": false

line added to them, which should resolve this unwanted darkening.

A comparison of potted objects with and without the fix can be found below:

22w11a (current rendering)

22w11a (fixed rendering)

Further notes

22w11a implemented potted mangrove propagules, which are completely unaffected by this issue affected slightly differently (see

MC-262696

Open

) as they have a unique, dedicated model. Despite this, shading was not disabled for existing potted crosses in this version.

MANUAL RESULT

Version used	
Reproduced?	YES / NO / PARTIAL
Crash file	
Notes	
Tested by / date	

Live issue: <https://bugs.mojang.com/browse/MC/issues/MC-129826>